

# Design review #4 – Space graded INA



|                       |                            |
|-----------------------|----------------------------|
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# 1 General information

This project will develop an instrumentation amplifier (INA) with programmable gain (through I2C / SPI interfaces) which achieves high accuracy (< 1% gain error), low noise (< 50nV/rtHz at 1kHz), and high common mode rejection (> 80dB). It will produce an application specific integrated circuit (ASIC) using free and open tools which will be taped out on IHP's 130nm open source PDK, with performance guaranteed across process (slow / fast silicon), voltage ( $\pm 5\%$ ) and temperature. Because of its high performance, the INA will be applicable to high accuracy sensing applications (pressure, strain, temperature) typical to those found in a wide variety of industry / medical applications. In addition, the INA will be made robust to high radiation environments, making it applicable to low earth orbit / high energy particle applications, where single even latch up can be problematic. The final chip will be characterised as a tangible proof of the maturity of today's free and open source toolchains and their readiness to produce high performance semiconductors.

## 1.1 Project goal

To show the capability of open source tools to develop industry quality ASICs, verified across industry standard conditions and taped out on silicon from a European fab (IHP).

To validate this goal, targeted performance will match that of an INA from a leading industry vendor (AD8223A [1]).

## 1.2 Purpose of Design review #4 (DR4)

Purpose of DR4 is to review the final ASIC before delivery to fab (i.e. tapeout) integration. Therefore, upon approval of DR4, ASIC is "frozen" and taped out.

## 1.3 Repository details

All material relating to this project can be found in the below publicly available repository:

[https://github.com/SLICESemiconductor/Space\\_grade\\_Instrumentation\\_Amplifier\\_ASIC/tree/main](https://github.com/SLICESemiconductor/Space_grade_Instrumentation_Amplifier_ASIC/tree/main)

Screenshot of the project landing page is shown in Figure 1.

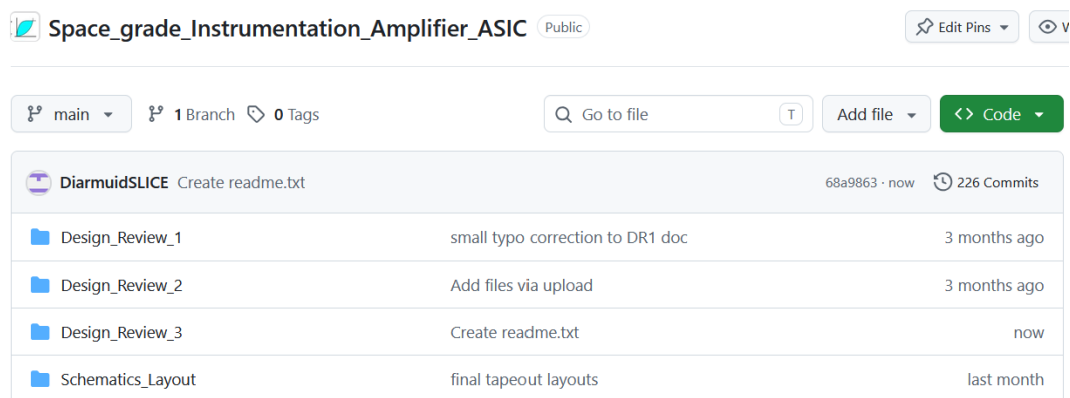


Figure 1 Project landing page

As this document pertains to DR4, all material relating to it is found in the sub directory "Design\_Review\_4". As block schematics relate to all DR's, these are contained in the separate folder "Schematics\_Layout". As shown in Figure 2 this folder contains a sub-directory to each block in the ASIC design.

Note: Figure 2 shows only a small portion of the total number of blocks contained in this folder.

Inside each folder here, lives the following views:

- Schematic (frozen at DR2)
- Symbol (frozen at DR2)
- Layout (frozen at DR3)

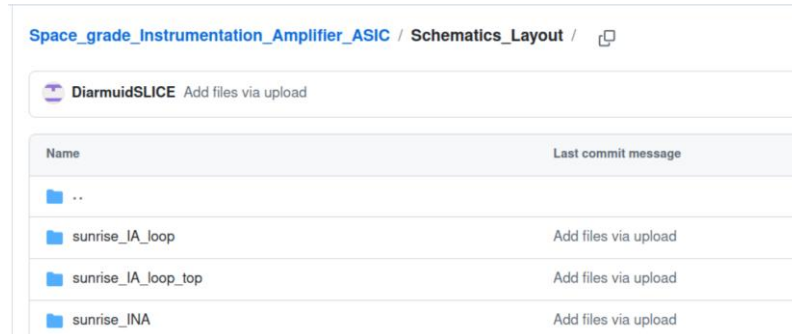


Figure 2 Project design blocks

## 1.4 Open source tools

The open source tools used for the ASIC are summarized in Table 2, with links to install each tool provided as references. In addition, tool versions used for the tapeout are included to enable results to replicated exactly at a later date.

| Tool    | Description                          | Version |
|---------|--------------------------------------|---------|
| ngspice | Spice simulator [2]                  | 42+     |
| xschem  | Schematic / waveform viewer [3]      | 3.4.8RC |
| openVAF | verilogA compiler [4]                | 23.5.0  |
| Octave  | Post processing [5]                  | 10.1.0  |
| klayout | Layout viewer / editor [6]           | 0.30.7  |
| magic   | Pcell generator / PEX generation [7] | 8.3.664 |
| netgen  | LVS [8]                              | 1.5.318 |

Table 1: Open source tools used for this ASIC

## 1.5 Open source PDK

The open source process development kit (PDK) used for the design is from IHP, process option – SG13CMOS5L [9]. This is a derivative of their more general option – SG13G2 [10] as it excludes the following:

- Top metal layer 2
- Linear caps (MiM or MoM)
- Deep Nwell (DNW)
- HBT devices

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The first 3 items impacted the design but not the 4<sup>th</sup> since the design is not RF. SG13CMOS5L option was chosen primarily because its tapeout date of 13/07/26 suited the project timeline. In addition, it is lower cost (max = €1.5k/mm<sup>2</sup> as opposed to €2.8k/mm<sup>2</sup> for SG13G2) and runs though the fab quicker (5 months instead of 6) to achieve a fab out date in 16/12/26 [11].

## 2 Pinlist

Full pinlist of the ASIC was extensively reviewed in [12] resulting in the list summarized in Table 2.

| #  | Name          | Description                                            | IO Type | Domain  | Value           |
|----|---------------|--------------------------------------------------------|---------|---------|-----------------|
| 1  | SPI_CPOL      | Mode select ( <i>or rdac_ctrl[1]</i> )                 | IP      | Digital | 0 / vdda_hv     |
| 2  | SPI_CPHA      | Mode select ( <i>or rdac_ctrl[0]</i> )                 | IP      | Digital | 0 / vdda_hv     |
| 3  | I2C_SCL       | I2C serial clock lin ( <i>or cdac_sel[4]</i> )         | IP      | Digital | 0 / vdda_hv     |
| 4  | I2C_SDA_O     | I2C data out                                           | TRI     | Digital | 0 / vdda_hv     |
| 5  | I2C_SDA_I     | I2C data in ( <i>or cdac_sel[3]</i> )                  | IP      | Digital | 0 / vdda_hv     |
| 6  | SPI_nCS       | SPI chip select ( <i>or INA pd</i> )                   | IP      | Digital | 0 / vdda_hv     |
| 7  | SPI_SCLK      | SPI serial clock line ( <i>or int bias</i> )           | IP      | Digital | 0 / vdda_hv     |
| 8  | SPI_MOSI      | SPI MOSI data line ( <i>or ip_ctrl[1]</i> )            | IP      | Digital | 0 / vdda_hv     |
| 9  | SPI_MISO      | SPI MISO data line                                     | TRI     | Digital | 0 / vdda_hv     |
| 10 | SPI_MISO_CTRL | SPI MISO tri-stat ctrl bypass ( <i>or ip_ctrl[0]</i> ) | IP      | Digital | 0 / vdda_hv     |
| 11 | CLK_EXT       | External clock                                         | IP      | Digital | 0 / vdda_hv     |
| 12 | RO_OUT        | RO output                                              | OP      | Digital | 0 / vdda_hv     |
| 13 | CLK_INT_EN    | Pass RO o/p to digital                                 | IP      | Digital | 0 / vdda_hv     |
| 14 | RO_EN         | Enable the RO                                          | IP      | Digital | 0 / vdda_hv     |
| 15 | VTSNS2_N      | Temp sensor bias                                       | ANA     | Analog  | 1 / 10uA        |
| 16 | VTSNS2_P      | Temp sensor bias                                       | ANA     | Analog  | 1 / 10uA        |
| 17 | VREF          | Voltage VOUT is referred to                            | ANA     | Analog  | 0 – Vref_max    |
| 18 | VOUT          | INA output                                             | ANA     | Analog  | 50mV – 3.3-50mV |
| 19 | VSSD          | Digital ground                                         | Supply  | Digital | 0V              |
| 20 | VDDA_LV       | Low voltage supply                                     | PWR     | Digital | 1.2 – 1.5V      |
| 21 | IBIAS_IN_5U   | External bias                                          | ANA     | Analog  | 5uA             |
| 22 | INT_BIAS      | Sense int_bias internal signal                         | OP      | Digital | 0 / vdda_hv     |
| 23 | PORB          | Power on reset (active lo)                             | IP      | Digital | 0 / vdda_hv     |
| 24 | SERIAL_SEL    | Select between SPI and I2C ( <i>or cdac_sel[5]</i> )   | IP      | Digital | 0 / vdda_hv     |
| 25 | TEST_MODE     | Repurpose SPI / I2C controls as direct i/p's           | IP      | Digital | 0 / vdda_hv     |
| 26 | VSSA          | Analog ground                                          | GND     | Analog  | 0V              |
| 27 | VDDA_HV       | High voltage supply                                    | PWR     | Analog  | 3.3V            |
| 28 | VSNS_EXT_N1   | External sense (-ve)                                   | ANA     | Analog  | 0.9 – Vcm_max_n |

| #  | Name        | Description                                   | IO Type | Domain  | Value           |
|----|-------------|-----------------------------------------------|---------|---------|-----------------|
| 29 | VSNS_EXT_P1 | External sense (-ve)                          | ANA     | Analog  | 0.9 – Vcm_max_n |
| 30 | VTSNS1_N    | Temp sensor bias                              | ANA     | Analog  | 1 / 10uA        |
| 31 | VTSNS1_P    | Temp sensor bias                              | ANA     | Analog  | 1 / 10uA        |
| 32 | VSNS_EXT_N2 | External sense (-ve) with 260Ω CDM protection | ANA     | Analog  | 0.9 – Vcm_max_n |
| 33 | VSNS_EXT_P2 | External sense (+ve) with 260Ω CDM protection | ANA     | Analog  | 0.9 – Vcm_max_p |
| 34 | I2C_ADDR[0] | Address (or cdac_sel[0])                      | IP      | Digital | 0 / vdda_hv     |
| 35 | I2C_ADDR[1] | Address (or cdac_sel[1])                      | IP      | Digital | 0 / vdda_hv     |
| 36 | I2C_ADDR[2] | Address (or cdac_sel[2])                      | IP      | Digital | 0 / vdda_hv     |

### 3 Package

Table 2 Top level pinlist

As shown in Table 2 a total of 36 pins are required. As a result, final ASIC will be delivered in a QFN36 package, provided by Serma [13] whose pin locations are shown in Figure 3.

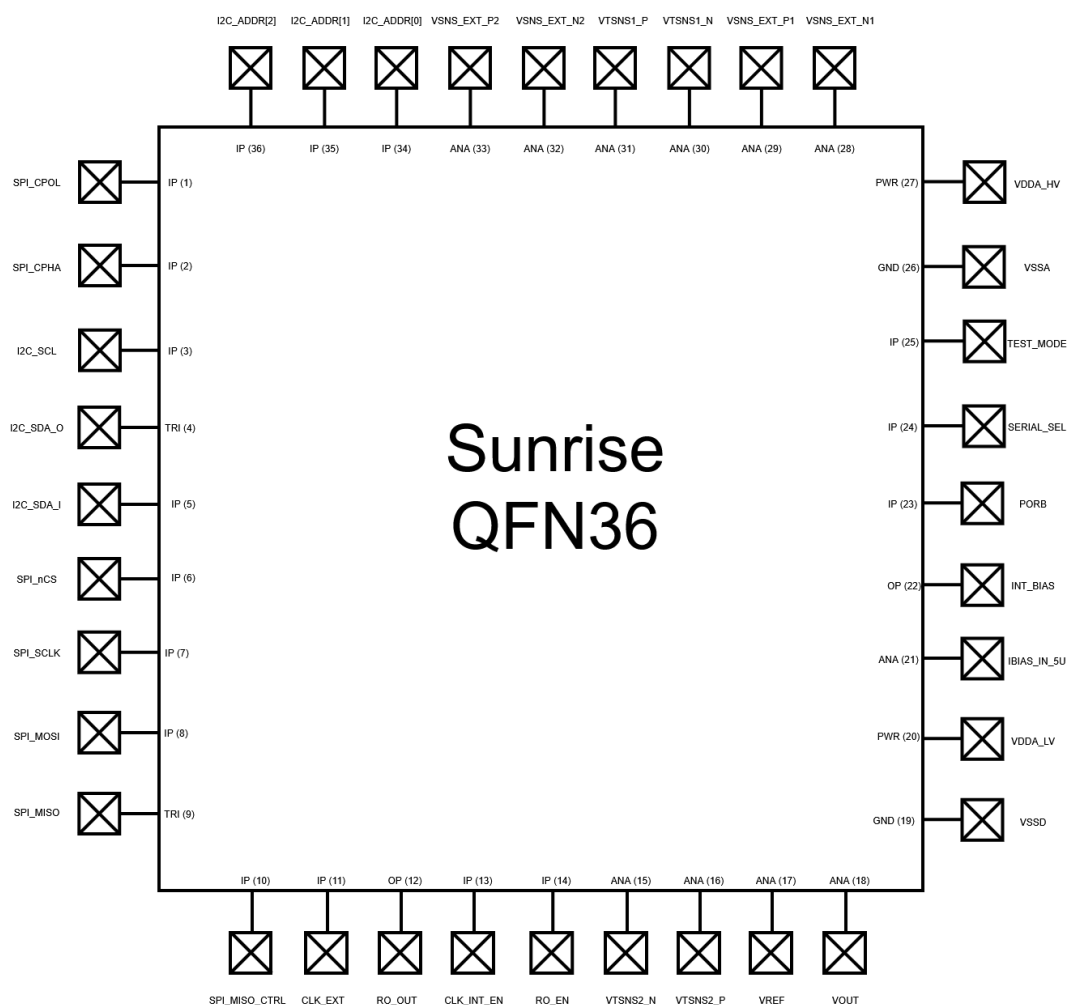


Figure 3 Sunrise pin placement

It should be noted that this pin placement drove the ASIC top level floorplan as shown in section 5.

## 4 Padding and seairing

### 4.1 Padding

All pins in Figure 3 must connect to internal IO cells which provide suitable ESD protection (typically 4kV HBM). As can be seen in Figure 3, a total of 8 types of IC cell are used, as summarised in Table 3 *Table* .

| # | Cell name               | Description       |
|---|-------------------------|-------------------|
| 1 | sunrise_pad_analog_io   | analog io         |
| 2 | sunrise_pad_digital_i   | digital input     |
| 3 | sunrise_pad_digital_o   | digital output    |
| 4 | sunrise_pad_digital_tri | digital tri state |
| 5 | sunrise_pad_pwr_iovdd   | io supply (3v3)   |
| 6 | sunrise_pad_pwr_iovss   | io gnd            |
| 7 | sunrise_pad_pwr_vdd     | core supply (1v5) |
| 8 | sunrise_pad_pwr_vss     | core gnd          |

Table 3 iocells used in sunrise

All the above cells are provided by IHP and can be downloaded from [14]. Documentation on the cells can then be found at [15]. To give a better understanding of how these cells are implemented in sunrise, the analog\_io and digital\_i cells are shown in Figure 4.

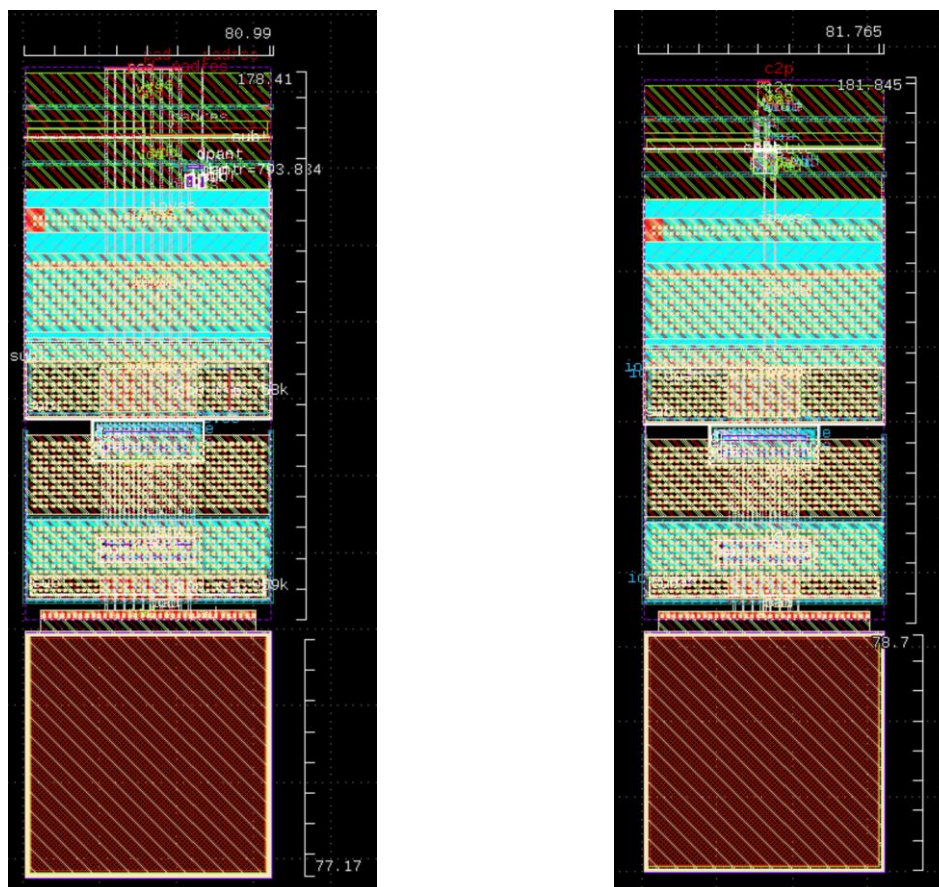


Figure 4 a) Analog io cell + bondpad

b) Digital input cell + bondpad

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In Figure 4, the IO cells are shown to occupy a footprint of  $\sim 80 \times 180 \mu\text{m}$ . Also shown is the  $80 \times 80 \mu\text{m}$  square bondpad connected to the IO cell which connects to an external pin.

Each of the 8 cells have been extracted and individually verified to be working as expected. For digital cells this meant toggling inputs and observing outputs, or vice versa. For analog cells it meant placing those cells on the relevant pins in the testbench to verify analog performance was not impacted.

Actual ESD robustness of the cells was verified in [16] for the S8 process, which also applies to 5L since the cells only differ by the top metal layer.

Finally the IO cells were connected to form a padding compatible with the pinout of Figure 3. Resulting padding is shown in Figure 5.

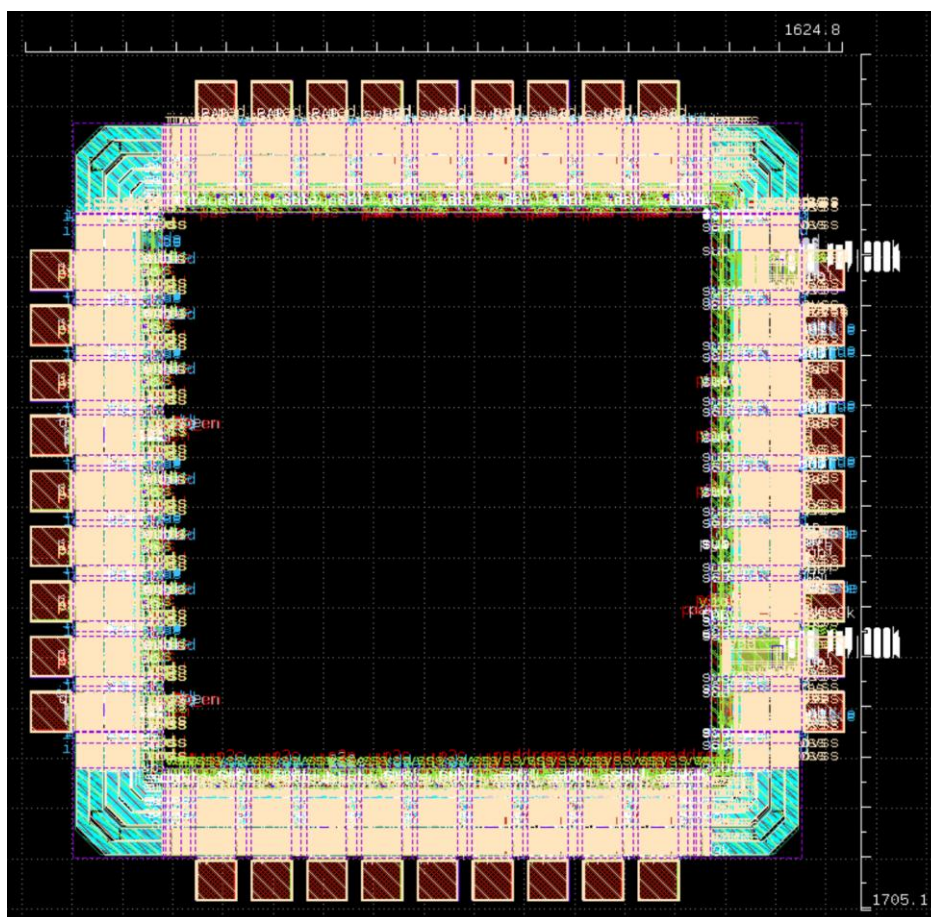


Figure 5 sunrise padding

It should be noted that when creating the padding, only the layouts of the IO cells were available, not their corresponding schematics. As a result, LVS could not be performed on the padding.

Since taping out, we have been informed by the JKU that such schematics are available at [17]. Therefore, post tapeout we plan to download these spice netlists, attach them to a symbolic view of the padding and perform LVS on that. There are various steps involved here so it will take some time however.

Not having LVS'd the padding is seen as low risk since it was closely inspected visually, multiple times, using the probe tool in klayout.

## 4.2 Seairng

The seairng defines the physical boundary of the die and thus defines the final dimensions of it. Seairng for sunrise is shown in Figure 6. Final dimensions taken from this seairng are shown in Figure 7 to be  $\sim 1.75 \times 1.75 \text{mm}$  to give a total area of  $\sim 3.1 \text{mm}^2$ .

|             |                       |      |       |                                                                                       |
|-------------|-----------------------|------|-------|---------------------------------------------------------------------------------------|
| Valid from: | Document ID:          | Rev. | Page: |  |
| 28.08.2026  | DR4: Space graded INA | A    | 7/19  |                                                                                       |

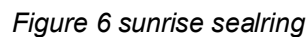
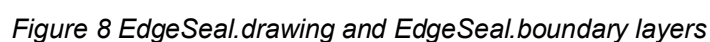


Figure 8 shows the two layers which unfortunately are given the same colour making them a little difficult to distinguish. Therefore, Figure 9 shows a close up of the two layers to better show their difference.



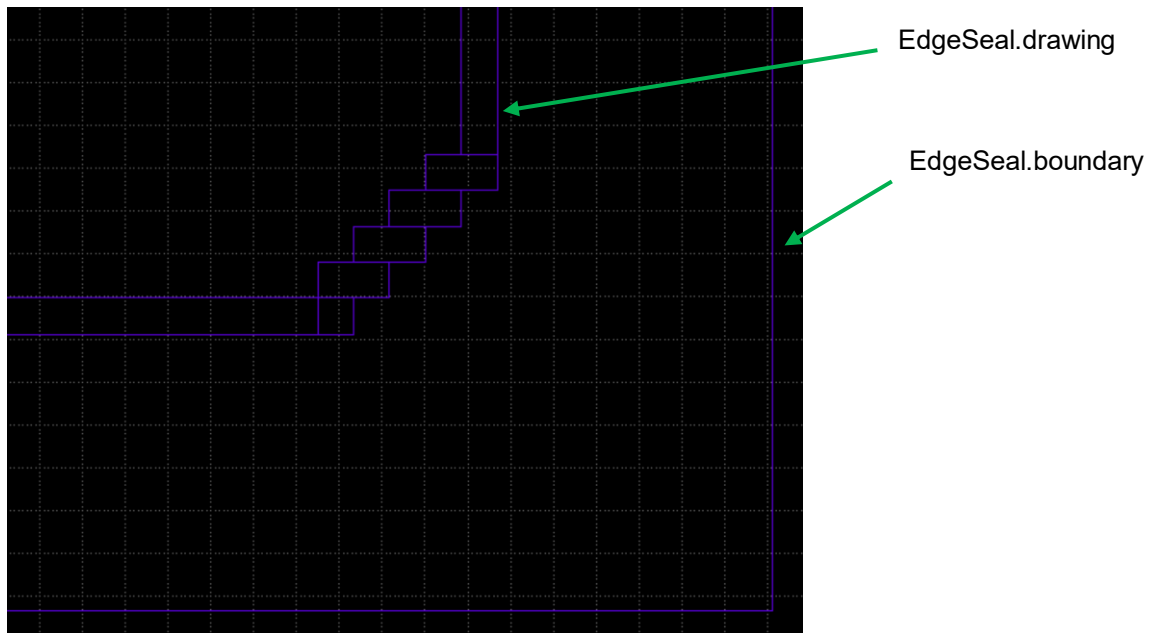


Figure 9 Close up of *EdgeSeal.drawing* and *EdgeSeal.boundary* layers

### 4.3 Pading and seal ring

For completion, Figure 10 shows the padding and sealing of sunrise combined.

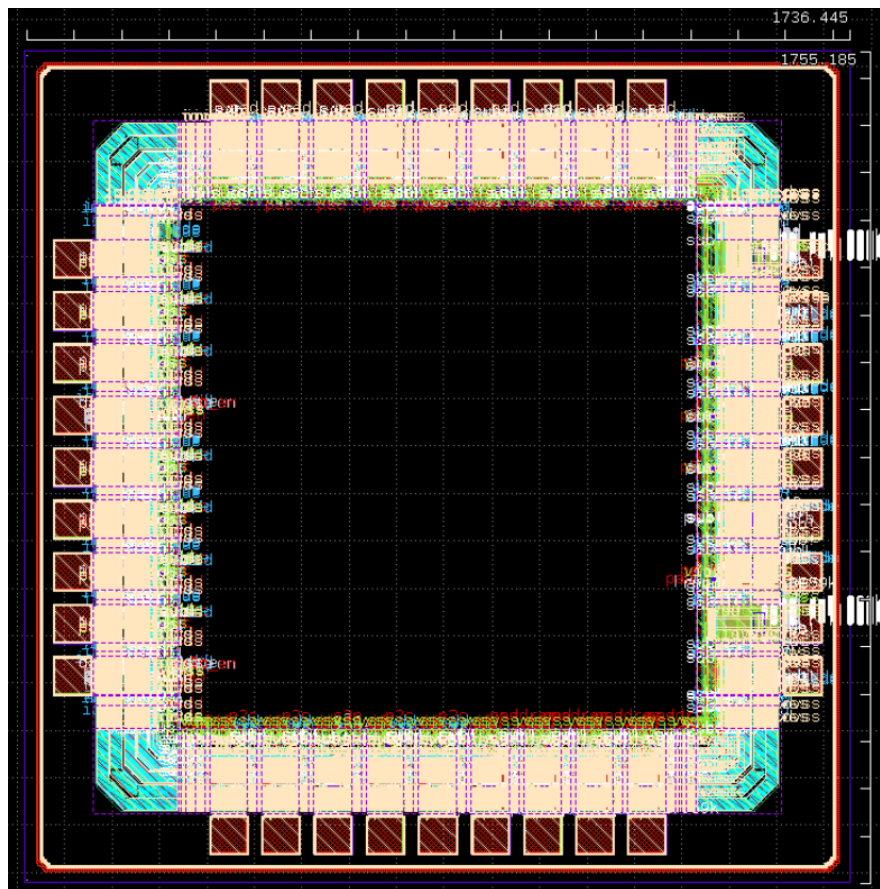


Figure 10 sunrise padding and sealing combined

## 5 Top level layout

Top level layout is shown in Figure 11 where it can be seen to occupy a footprint of  $\sim 1 \times 1 \text{ mm}$ . Comparing with the pin placement of Figure 3 one can see how that pin placement drove the top level floorplan (e.g ring oscillator sits in the bottom left corner close to all its control pins, main temp sensor sits in the top center close to its inputs, digital sits towards the left hand side close to its control pins, etc).

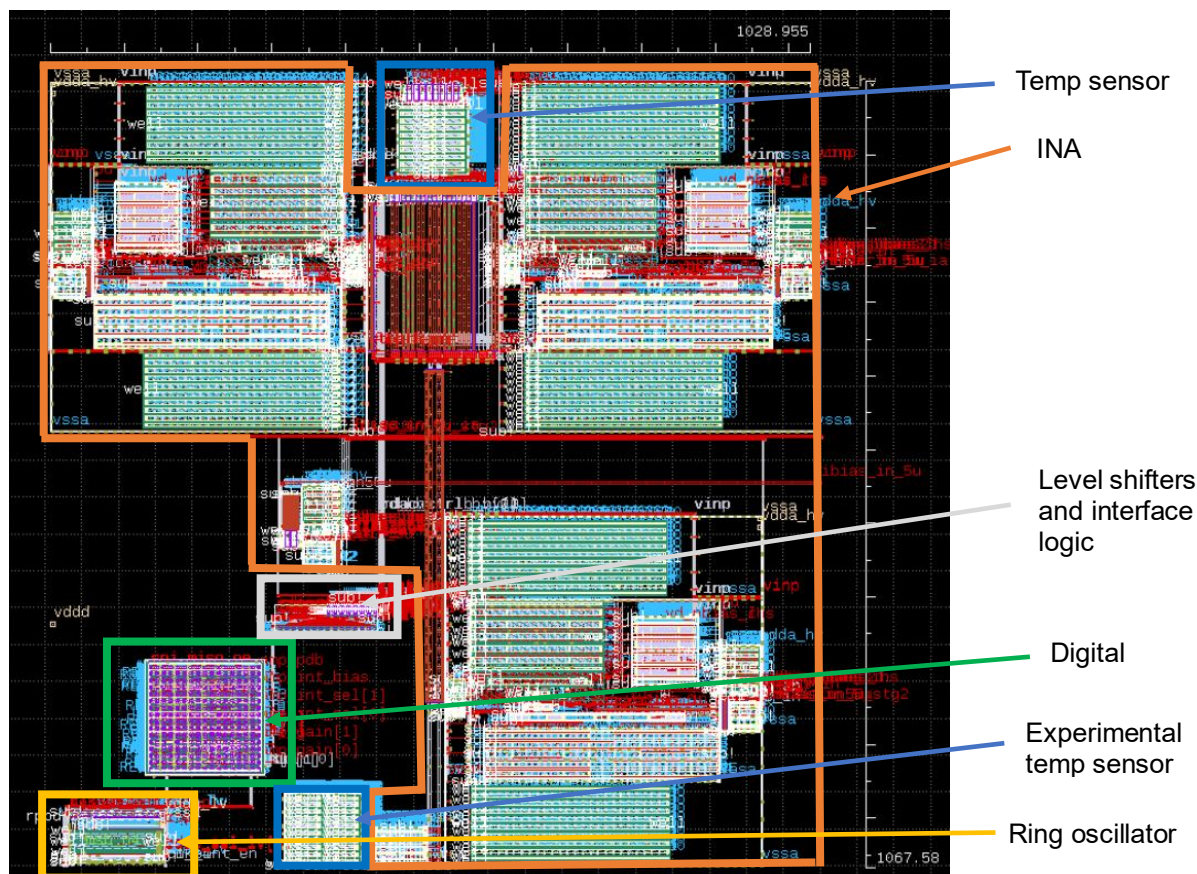


Figure 11 sunrise top level layout

A schematic view instantiating all the above blocks, apart from the digital was created and LVS'd against the corresponding layout. Performing LVS on a view including the digital is a difficult task, even in commercial tools. An approach has been proposed which we plan to investigate post layout.

Not having LVS'd the top level including the digital is seen as low risk since it was closely inspected visually, multiple times, using the probe tool in klayout.

It is hoped that the digital and padding post tapeout LVS activities will be successful, as the resulting flow will enable full chips for future projects to be LVS'd in their entirety, thus providing a very thorough signoff.

Top level layout including the power grid of M4/M5 is shown in Figure 12.



Micrograph showing a device registration grid. The central area is a green, textured chip. The grid consists of a series of small, dark, rectangular registration marks arranged in a rectangular pattern around the chip. The grid is labeled with 'X' and 'Y' coordinates. The chip area is labeled with 'X' and 'Y' coordinates. The grid is labeled with 'X' and 'Y' coordinates. The chip area is labeled with 'X' and 'Y' coordinates.

Device registration size:  $x=1740.8\text{ }\mu\text{m}$ ,  $y=1754.8\text{ }\mu\text{m}$   
 Calculated area:  $3.1\text{e-06 sq mm}$

Figure 13 sunrise top level layout with power grid, padding and searing

## 6 Dummy fill generation

To avoid over / under etching during the CMP steps of the fabrication process, all metal / poly / active densities must lie within a range specified by the foundry. To achieve this, dummy fill is generated and placed between all specified areas. Fill generation script is provided by the foundry which must not overfill creating max density violations, whilst at the same time place enough fill to avoid min density violations.

As dummy fill can affect matching, exclusion zones are placed above matching sensitive analog circuitry where the fill algorithm knows not to place any fill. An example of this is shown in Figure 14 where M1 fill has been applied in all required spaces except above the matching critical analog circuit which has been defined by the M1 fill block (Metal.nofill).

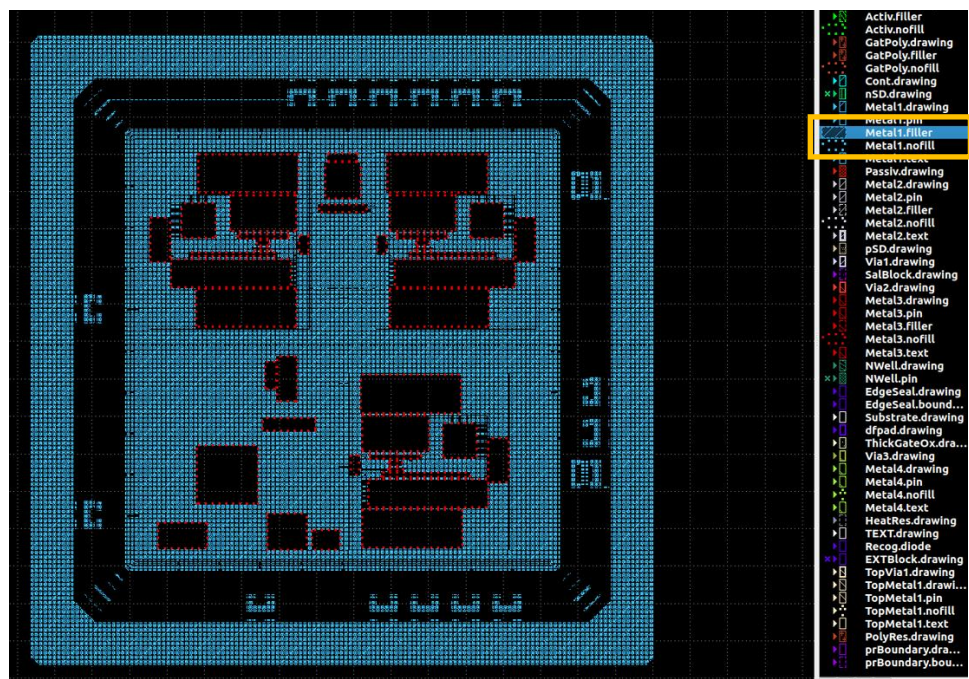


Figure 14 M1 fill

Metal fill spacings were specified as per Figure 15.

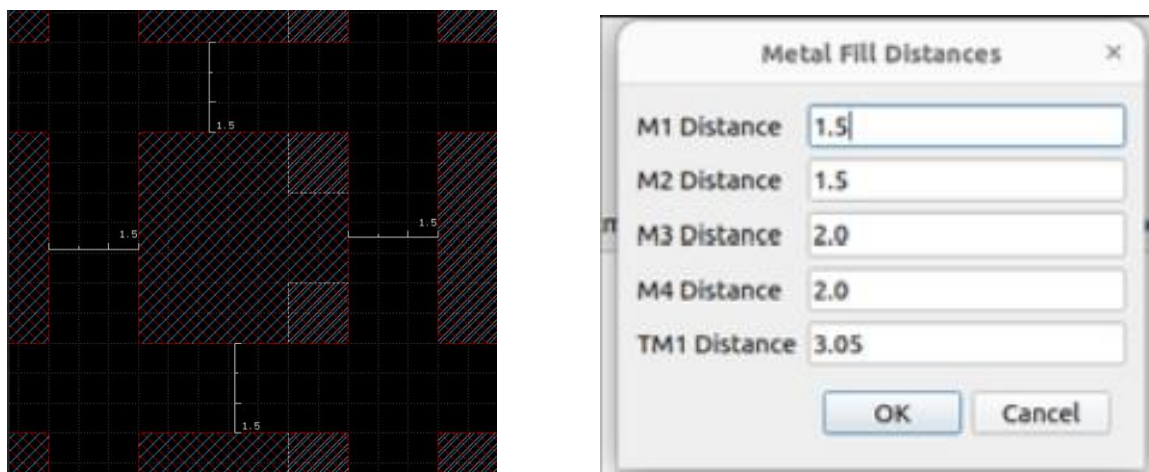


Figure 15 Metal fill spacings

Resulting fill of all the layers at the top level is shown in Figure 16, which represents the final top level view of the sunrise ASIC.

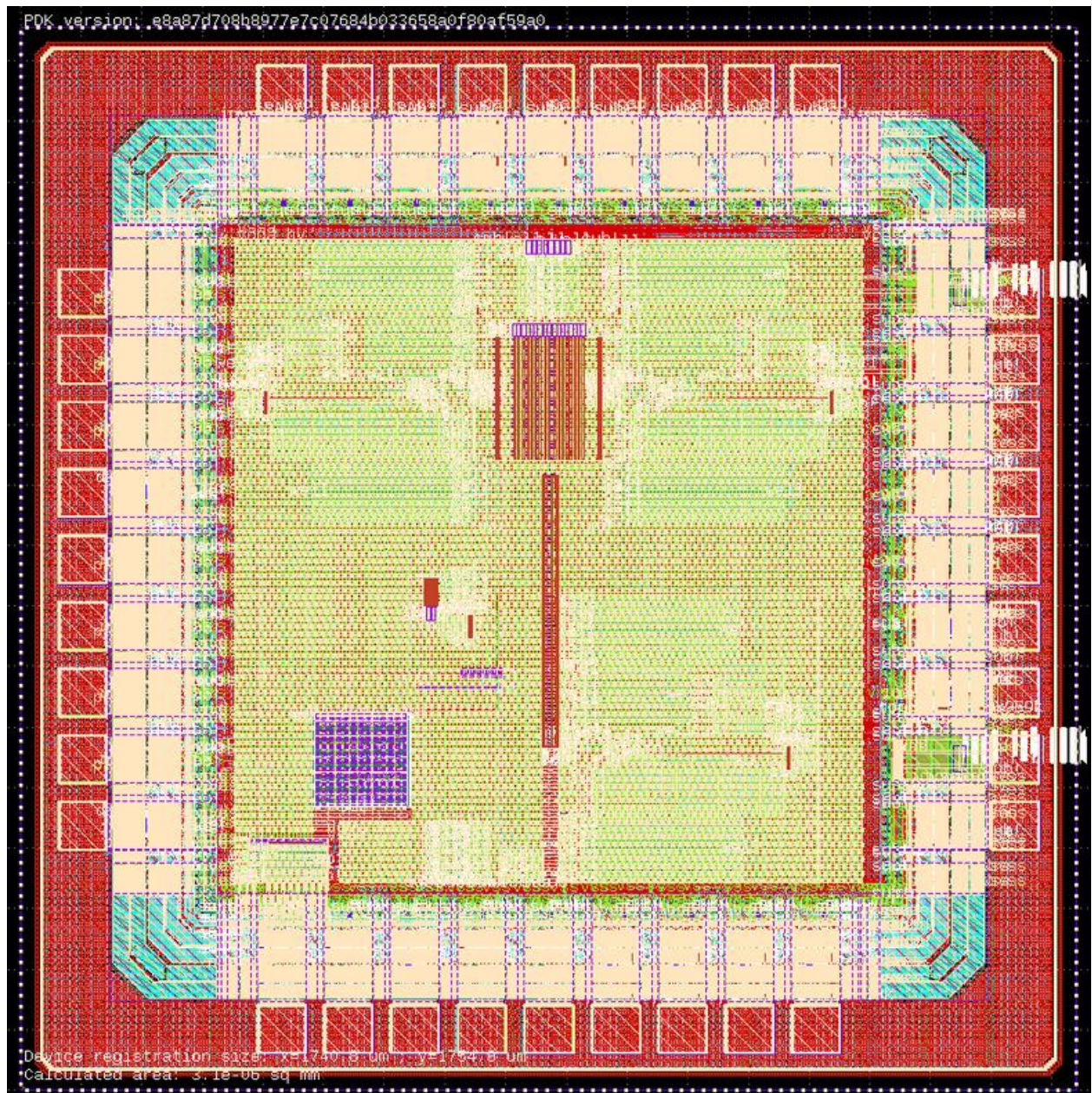


Figure 16 sunrise final ASIC

Note in Figure 16 how no fill is applied between the EdgeSeal.drawing and EdgeSeal.boundary layers since this region is left blank because of the die cutting process as described in section 4.

Prior to tape out, DRC density check must be run on the top level layout. DRC density checker from IHP was found to have the below bugs:

- Included up to the EdgeSeal.boundary and not the EdgeSeal.drawing
- Double counted (e.g. treated 2 identical metals placed on top of each other as 2 separate metals)

These bugs resulted in incorrect density results. In particular, the 2<sup>nd</sup> bug resulted in density numbers  $\gg 100\%$  as shown in Figure 17.

```

Layer areas and percentages (units: um^2)
-----
Total Area      = 3856.31 um^2

Activ   Area      = 1476664.54 um^2
Activ   Density % = 38292.14% [Min = 35.00%, Max =

GatPoly Area      = 577226.20 um^2
GatPoly Density % = 14968.34% [Min = 15.00%]

Metal1  Area      = 1418750.71 um^2
Metal1  Density % = 36790.34% [Min = 35.00%, Max =

Metal2  Area      = 1126023.13 um^2
Metal2  Density % = 29199.48% [Min = 35.00%, Max =

```

Figure 17 Incorrect density calculations from the IHP script

To address this, Yacine Sotehi investigated and fixed these bugs. Re-running the top level layout using this updated script showed much more reasonable results, provided in Figure 18.

```

#####
:          Density Report          #
: #####
:
: Layer areas and percentages (units: um^2)
: -----
:
: Total Area      = 2833080.96 um^2
:
: Activ   Area      = 1476664.54 um^2
: Activ   Density % = 52.12% [Min = 35.00%, Max = 55.00%]
:
: GatPoly Area      = 577226.20 um^2
: GatPoly Density % = 20.37% [Min = 15.00%]
:
: Metal1  Area      = 1418750.71 um^2
: Metal1  Density % = 50.08% [Min = 35.00%, Max = 60.00%]
:
: Metal2  Area      = 1125930.58 um^2
: Metal2  Density % = 39.74% [Min = 35.00%, Max = 60.00%]
:
: Metal3  Area      = 1444605.59 um^2
: Metal3  Density % = 50.99% [Min = 35.00%, Max = 60.00%]
:
: Metal4  Area      = 1326832.28 um^2
: Metal4  Density % = 46.83% [Min = 35.00%, Max = 60.00%]
:
: TopMetal1 Area      = 1396418.82 um^2
: TopMetal1 Density % = 49.29% [Min = 25.00%, Max = 70.00%]
:
:

```

Figure 18 Correct density calculations from the updated IHP script

These results were verified as they matched those provided by calibre (the gold standard for pre-tapeout checks).

To use this updated, the user must do the following:

- Go to your \$PDK\_ROOT area
- git clone <https://github.com/YacineSot/ihp-sg13cmos5l> <some\_name>
- Set \$PDK to <some\_name>
- Run DRC density check as normal

## 7 Final DRC sign off

DRC of the entire top level ASIC had to be performed in magic since it kept crashing klayout. Suspected reason is the DRC deck rules are coded much more efficiently in magic thus requiring less memory to execute. Resulting top level DRC results are shown in Figure 19.

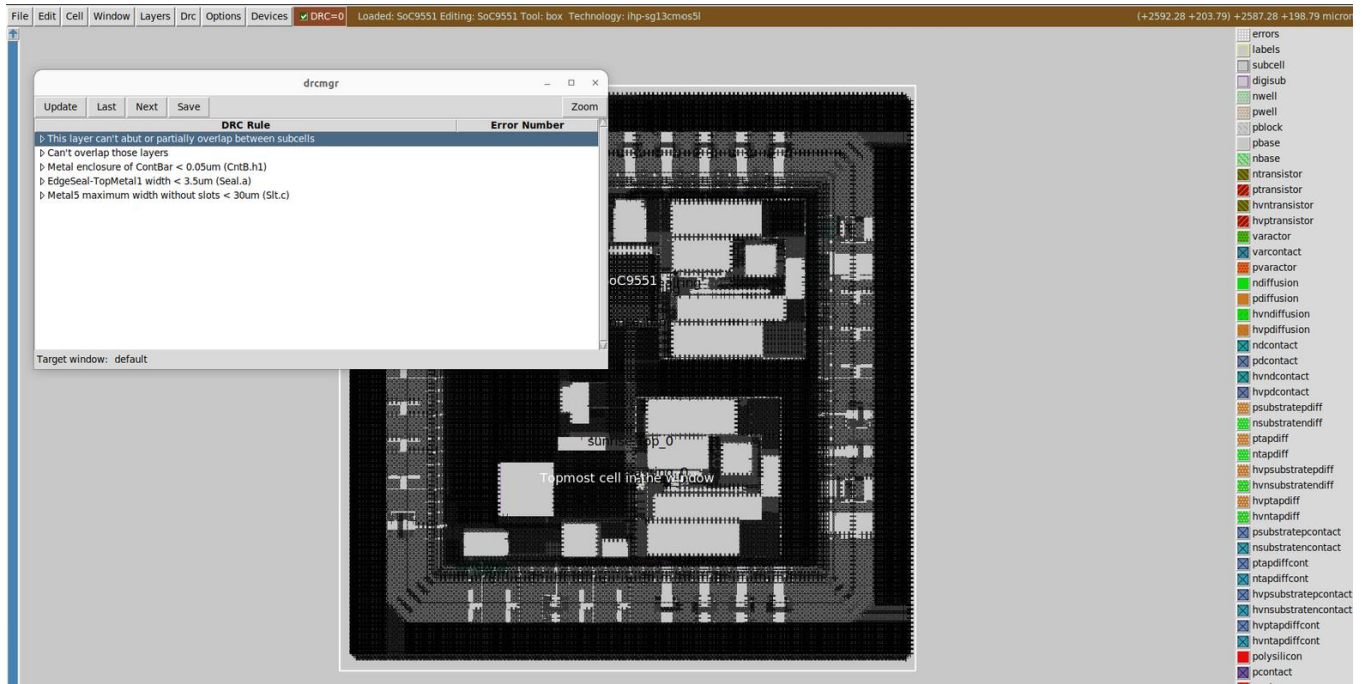


Figure 19 sunrise DRC signoff results from magic

A total of 5 DRC categories were reported. All were investigated and confirmed by IHP / magic tool owner (Tim Edwards) to be either waivable or due to a bug in the DRC checker. Therefore, the above constitutes a clean DRC result. Calibre (industry golden reference sign off for DRC) was next applied by IHP to the top level pre-tape out, which verified this clean result.

## 8 Final top level simulations

Although all blocks were simulated extensively post layout in their respective DR3's, as a final sanity check, top level specs were simulated at TT/27C on an extracted netlist containing all analog blocks. Results are summarized in Table 4.

|                 | SCH      | RCX      |
|-----------------|----------|----------|
| CMRR (dB)       | -114.372 | -114.784 |
| GE (%)          | -0.117   | -0.117   |
| fosc (MHz)      | 10.45    | 9.82     |
| D_osc (%)       | 52.48    | 51.02    |
| idda (uA)       | 785.7    | 785.6    |
| BW (kHz)        | 301.4    | 300.1    |
| Terror (degC)   | 0.697    | 0.699    |
| vout_sample (V) | 1.7      | 1.7      |

Table 4 Final top level simulation results pre (SCH) and post (RCX) layout

As shown in Table 4, pre and post layout results match very well to completely verify top level performance and hook up.

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It should be noted that the digital block was left out of this analysis. This is because the co-sim environment to simultaneously test spice and Verilog netlists was not set up pre-tapeout. Although an involved task, this will be investigated post tapeout since it represents a gap in the verification. This gap is deemed low risk since the Verilog has been thoroughly verified pre and post silicon with its hook up to the analog being closely inspected visually, multiple times, using the probe tool in layout.

## 9 Final tapeout submission

Final tapeout was committed to the below repo [18] as requested by IHP.

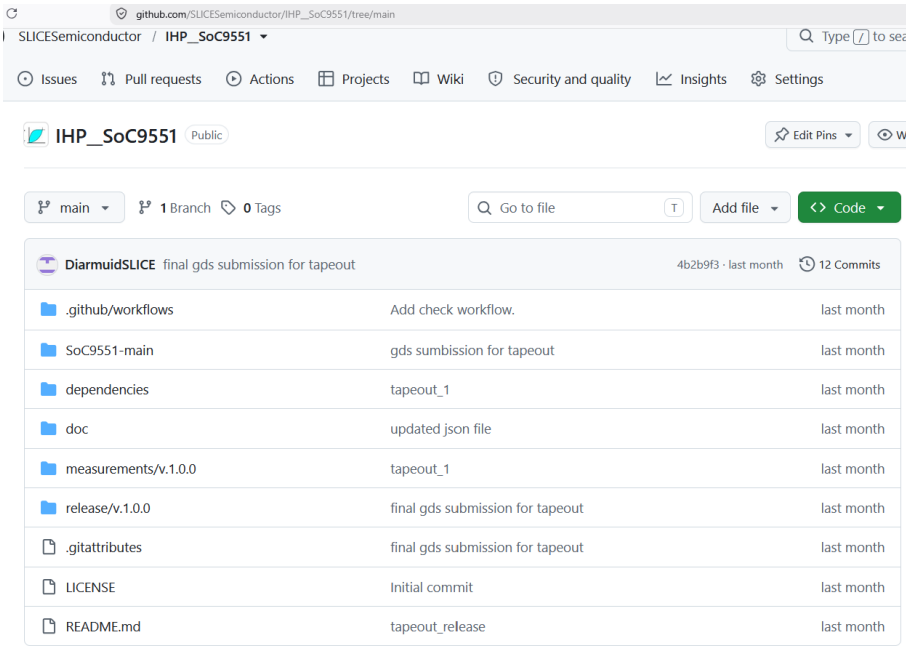


Figure 20 Sunrise tapeout repo at IHP

Tapeout requested was then submitted at [19] where all tapeout checks were completed / signed off on 16.07.26 as shown below.

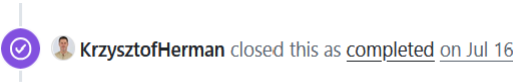


Figure 21 Final tapeout confirmation from IHP

## 10 Post silicon lab characterization

Table 5 summarises the equipment required to characterize this part post silicon. It should be noted that at the time of writing, all required equipment apart from the oscilloscope, have been acquired. We are currently investigating the most suitable oscilloscope for the best price.

| Equipment    |                    |                 |
|--------------|--------------------|-----------------|
| Abbreviation | Description        | Model           |
| PSU          | Power supply unit  | Agilent E3631A  |
| SMU          | Source meter unit  | Keithley 2602B  |
| DMM          | Digital multimeter | Agilent 34970A  |
| SIGGEN       | Signal generator   | Keysight 33500B |
| SPEC         | Spectrum analyser  | Anritsu MS2691A |
| OSC          | Oscilloscope       | tbd             |
| TC           | Thermocouple       | K-type          |

Table 5 Lab char equipment list

Building on Table 5, Table 6 summarises all the specs which will be characterized post silicon. It provides a brief description of how each spec will be measured, equipment used to make that measurement and char board requirements to facilitate the measurement.

| Spec       | PSU | SMU | DMM | SIGGEN | SPEC | OSC | TC | Char board requirements                                                 | Brief measurement description                                                                                                                                                                                                         |
|------------|-----|-----|-----|--------|------|-----|----|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GE         | Y   | Y   | Y   | Y      | N    | N   | N  | Sense pts at INA i/ps                                                   | Measure output for various discrete diff i/ps + also a ramped i/p                                                                                                                                                                     |
| CMRR       | Y   | Y   | Y   | N      | N    | N   | N  | Sense pts at INA i/ps                                                   | Measure output for a given diff i/p at different common mode levels                                                                                                                                                                   |
| Noise      | Y   | Y   | Y   | N      | Y    | N   | N  | Low noise BNC at INA o/p                                                | Measure noise density with INA gain set to 21                                                                                                                                                                                         |
| CSA        | Y   | Y   | Y   | N      | N    | N   | N  | Resistor dividers (x2) which can be connected to INA i/ps using jumpers | Keithley sources current through resistor divider ( $r_{top} = r_{sense}$ ) which sends a diff voltage into the INA at a specified common mode level. Measure o/p and correlate result with sense resistor.                           |
| fosc       | Y   | N   | N   | N      | Y    | Y   | N  | BNC at RO o/p                                                           | Measure fosc using oscilloscope and spectrum analyser                                                                                                                                                                                 |
| fosc res   | Y   | N   | N   | N      | Y    | Y   | N  | BNC at RO o/p                                                           | Measure fosc for trim code increments                                                                                                                                                                                                 |
| trim range | Y   | N   | N   | N      | Y    | Y   | N  | BNC at RO o/p                                                           | Measure fosc at min and max trim codes                                                                                                                                                                                                |
| duty cycle | Y   | N   | N   | N      | N    | Y   | N  | BNC at RO o/p                                                           | Measure duty cycle on the OSC                                                                                                                                                                                                         |
| Tsns       | Y   | Y   | Y   | N      | N    | N   | Y  | Sense pts at Tsns i/ps<br>Automated switching of i/p's before sense pts | Measure dvgs and using the simulated $T(vgs)$ transfer function, back calculate the junction temperature ( $T_j$ ). Correlate with thermocouple ambient measurement ( $T_a$ ), accounting for self-heating ( $T_j = T_a - P.R_{th}$ ) |
| Power      | Y   | Y   | Y   | N      | N    | Y   | Y  | Sense pts at supply inputs                                              | Power up ASIC and read current from PSU                                                                                                                                                                                               |

Table 6 Post silicon char plan

Color coding scheme for Table 6 is explained in Table 7.

| Legend                  |
|-------------------------|
| INA related measurement |
| RO related measurement  |
| TS related measurement  |
| Full ASIC measurement   |

Table 7 Legend for Table 6

It should be noted that the above focuses only on characterizing the analog performance of the ASIC. As the ASIC is programmable, the char board needs to include a microcontroller which can interface with the I2C/SPI. In addition, dip switches must also be included to enable the user to manually code the INA as a risk mitigation in the event of the I2C/SPI not functioning as expected. Finally, to facilitate bringing an external clock on board, a BNC connector must be included and connected to pin 11 in Figure 3 (CLK\_EXT).

## 11 Power up procedure

The full digital to analog (D/A) interfacing of the ASIC is detailed in [20] which includes instructions on the correct sequence to power up the part. As this relates to lab char it is summarized here.

As detailed in [21], the level shifter was imported as IP which does not contain a reset. This introduces a risk whereby if vdda\_lv comes up before vdda\_hv, the level shifter could enter an undefined state leading to unwanted paths. To mitigate this it is vital the correct power up sequence is employed to guarantee vdda\_lv comes up before vdda\_hv. This is shown in Figure 22 and summarized in Table 8.

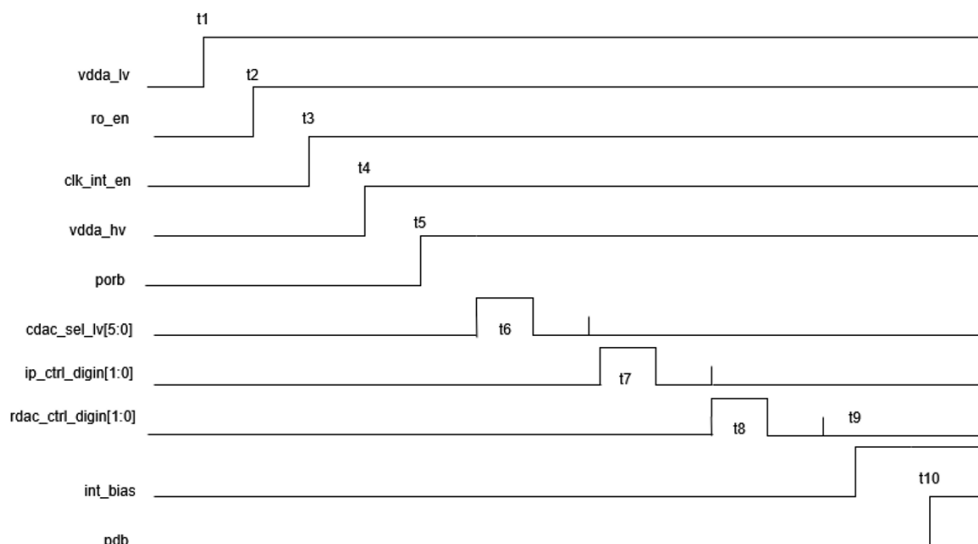


Figure 22 Power up sequence waveforms

| Time point | What happens                 | Description                                                                | Comment                                                                          |
|------------|------------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| t1         | vdda_lv ramps                | Bring up the digital supply                                                |                                                                                  |
| t2         | ro_en asserts                | RO starts oscillating                                                      |                                                                                  |
| t3         | clk_int_en asserts           | RO passes to the output / digital                                          | If RO is not working, apply an external clk and set clk_int_en = 0               |
| t4         | vdda_hv ramps                | Bring up the analog supply                                                 | Vdda_hv must come up after vdda_lv as level shifters have no reset               |
| t5         | porb asserts                 | Digital comes out of power on reset                                        | Up to this point all digital o/p's = 0                                           |
| t6         | Program cdac_sel_lv[5:0]     | Tune the RO 10MHz                                                          | Programmed through SPI / I2C                                                     |
| t7         | Program ip_ctrl_digin[1:0]   | Select the input to be sensed by the INA                                   | Programmed through SPI / I2C                                                     |
| t8         | Program rdac_ctrl_digin[1:0] | Set the gain of the INA                                                    | Programmed through SPI / I2C                                                     |
| t9         | Set int bias                 | If using internal bias, int_bias = 1, if using external bias, int_bias = 0 | External current source must only be turned on after int_bias = 0, or not at all |
| t10        | Assert pdb                   | Take the analog out of reset                                               | Since bias has been applied and all config bits set, analog can now be used      |

Table 8 Power up sequence summary

## 12 References

- [1] <https://www.analog.com/media/en/technical-documentation/data-sheets/AD8223.pdf>
- [2] <https://ngspice.sourceforge.io/download.html>
- [3] [https://xschem.sourceforge.io/stefan/xschem\\_man/install\\_xschem.html](https://xschem.sourceforge.io/stefan/xschem_man/install_xschem.html)
- [4] <https://openvaf.semimod.de/>
- [5] <https://octave.org/download>
- [6] <https://www.klayout.de/build.html>
- [7] <https://opencircuitdesign.com/magic/download.html>
- [8] <https://opencircuitdesign.com/netgen/>
- [9] <https://github.com/IHP-GmbH/ihp-sg13cmos5l>
- [10] <https://github.com/IHP-GmbH/IHP-Open-PDK>
- [11] <https://www.ihp-microelectronics.com/services/research-and-prototyping-service/mpw-prototyping-service/schedule-price-list>
- [12] [https://github.com/SLICESemiconductor/Space\\_grade\\_Instrumentation\\_Amplifier\\_ASIC/blob/main/Design\\_Review\\_4/SUNRISE\\_Pinlist\\_Padring.docx](https://github.com/SLICESemiconductor/Space_grade_Instrumentation_Amplifier_ASIC/blob/main/Design_Review_4/SUNRISE_Pinlist_Padring.docx)
- [13] <https://www.serma-microelectronics.com/en/>
- [14] [https://github.com/IHP-GmbH/ihp-sg13cmos5l/tree/main/libs.ref/sg13cmos5l\\_io/gds](https://github.com/IHP-GmbH/ihp-sg13cmos5l/tree/main/libs.ref/sg13cmos5l_io/gds)
- [15] [https://github.com/IHP-GmbH/IHP-Open-PDK/tree/dev/ihp-sg13g2/libs.ref/sg13g2\\_io/doc](https://github.com/IHP-GmbH/IHP-Open-PDK/tree/dev/ihp-sg13g2/libs.ref/sg13g2_io/doc)
- [16] [https://github.com/SLICESemiconductor/Space\\_grade\\_Instrumentation\\_Amplifier\\_ASIC/blob/main/Design\\_Review\\_4/Sunrise\\_padring.pdf](https://github.com/SLICESemiconductor/Space_grade_Instrumentation_Amplifier_ASIC/blob/main/Design_Review_4/Sunrise_padring.pdf)
- [17] [https://github.com/iic-jku/TinyWhisper/blob/main/ihp130/verification/lvs/sg13g2\\_io.spi](https://github.com/iic-jku/TinyWhisper/blob/main/ihp130/verification/lvs/sg13g2_io.spi)
- [18] [https://github.com/SLICESemiconductor/IHP\\_SoC9551/tree/main](https://github.com/SLICESemiconductor/IHP_SoC9551/tree/main)
- [19] <https://github.com/IHP-GmbH/Open-Silicon-MPW/issues/41#issuecomment-4955555848>
- [20] [https://github.com/SLICESemiconductor/Space\\_grade\\_Instrumentation\\_Amplifier\\_ASIC/blob/main/Design\\_Review\\_4/SUNRISE\\_DA\\_INTERFACE\\_RevB.docx](https://github.com/SLICESemiconductor/Space_grade_Instrumentation_Amplifier_ASIC/blob/main/Design_Review_4/SUNRISE_DA_INTERFACE_RevB.docx)
- [21] [https://github.com/SLICESemiconductor/Space\\_grade\\_Instrumentation\\_Amplifier\\_ASIC/tree/main/Design\\_Review\\_3/LVL\\_SHFT](https://github.com/SLICESemiconductor/Space_grade_Instrumentation_Amplifier_ASIC/tree/main/Design_Review_3/LVL_SHFT)

## 13 Revision History

| Rev. | Date       | Author           | Changes, Remarks |
|------|------------|------------------|------------------|
| A    | 08.08.2026 | Diarmuid Collins | Created          |
|      |            |                  |                  |
|      |            |                  |                  |

Table 9: Revision history